Tactix: DSR Checker-Contract Cross-Check

Report to Andrew

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Outcome

AM407 compared the historical no-option DSR checker route with `-delete-implied-units` on four archived, real Tactix CNF/proof pairs. All seven predeclared outcomes matched; the observation-only eighth also verified. One proof verifies only with the option; another verifies only without it; a known Sweep-retention regression is rejected by both; and its repaired exact case verifies under both. This is stronger contract evidence than hand-built controls alone, but neither mode has been promoted or proved universally sound.

Verification

Job `1721174` completed `0:0` in three seconds on hwulab with two CPUs and 8 GiB. Both modes used the same frozen AM369 `dsr-trim`, `dsr-compress` and Trestle pins. A pass required backward elaboration, independent Trestle verification, and then forward every-lemma elaboration and Trestle verification. The AM296 direct-SR proof under no-option shows why the independent stage matters: backward elaboration produces an LSR, but Trestle reports `s INVALID`. The opposite AM296 proof under the option fails at its proof line 13 because the requested deletion removes an implied-unit reason it later needs.

The exact runner, eight original inputs, per-stage logs, resulting LSRs and SHA-256 manifests are frozen in the `am407-dsr-archived-contract` probe under `benchmarks/probes/`. The AM407 technical report in `benchmarks/reports/` records the eight-row comparison and limits. No solver, production oracle, checker binary, or public API changed.

Decision and remaining work

These results make the choice consequential: the option is not a monotone relaxation. The canonical no-option route remains unchanged while the user decides whether to adopt a fully audited option contract or require a general checker-visible proof-state repair. H2 still fails no-option; post-finalization public clause addition can still produce a wrong SAT model. The Opus-owned AM400/401 fuzz legs and two full Debug suites are in flight. N1–N4, the retained Rank/B+C/WV gates, and production online-learning readiness are not complete. The requested full Tier-D comparison at 5,000 seconds, four CPUs and 32 GiB on hwulab is therefore not submitted.